Vijay Mathew

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Professional Summary

Software architect and language/runtime engineer with 20+ years of experience building programming languages, distributed services, and developer platforms. Works AI-natively: directs coding agents (Claude Code, Codex) to deliver production-scale systems, and keeps their output trustworthy through executable specifications, design-by-contract, written design reviews, and large automated test suites. Currently building durable, multi-actor runtimes for AI Employees at Fractl; creator of Nex, a contract-first programming language for verifiable software.

Core Strengths

- **AI-native engineering:** agentic coding workflows (Claude Code, Codex); spec- and contract-driven development; agent-readable codebases (coding conventions for agents, design → plan → gap review → disposition loops); verifying generated code at scale
- **Agent platforms:** LLM agent loops, durable execution (replay, recovery, outbox, idempotency), human-in-the-loop approvals, RBAC and tenant isolation for AI actors
- **Systems:** language design, compilers and runtimes, distributed systems, service architecture, developer tooling, search and content platforms
- **Languages and tools:** Clojure, Java, TypeScript, Scheme, C, Python; PostgreSQL; Langium, ANTLR

Professional Experience

Fractl

Apr 2018 – Present

Founding Engineer and Software Architect

Remote

- Architect and core engineer of AgentLang Next, Fractl's durable-execution framework that turns business processes into running AI Employee applications: language, compiler, runtime, server, CLI, and agent loop.
- Proposed and led the move from a custom DSL to an embedded TypeScript SDK: wrote the design, saw it adopted as the platform's only authoring language, and implemented the new compiler, model API, and CLI path.
- Wrote 160+ of the project's design, plan, and gap-review documents, which keep agent-generated changes architecturally coherent.
- Contributed to durable multi-actor process execution, RBAC and tenant-aware access control, and the agent turn loop that AI Employees run in.
- Earlier, led the design of Fractl's original declarative language and runtime for domain modeling.

SAP Concur

Dec 2014 – Apr 2018

Principal Engineer

Bangalore

- Led the design and implementation of a scalable identity management and authorization service.
- Built and brought the service to production as the primary engineer; it served 40+ million requests per day at launch.
- Stack: Clojure, Java, Python, MySQL, Elasticsearch, Couchbase.

HappiestMinds

Jun 2012 – Dec 2014

Senior Technical Lead

Bangalore

- Designed and built content indexing, search, and delivery services for Pearson Education's next-generation EPUB reader.
- Implemented the Python SDK for authentication services.
- Stack: Java, Spring REST, Python, Elasticsearch.

Actiance

Apr 2007 – Jun 2012

Senior Systems Analyst

Bangalore

- Designed the information leak prevention subsystem for an enterprise network monitoring platform, including content classification and in-stream policy enforcement.
- Built an RPC framework based on Google Protocol Buffers for efficient transport of captured IM/UC content.
- Delivered network auditing tools and integrations across IBM Sametime, Microsoft SharePoint, and Microsoft OCS; also built a Flex-based visualization tool for communication pattern analysis.

Earlier Experience

Jan 2000 – Apr 2007

SpikeSource, Mindteck, Fischer Systems, DArts, Indusware Solutions, Freelance

Bangalore, Trivandrum, Tiruvalla

- Built synchronization and integration products: a generic sync platform, an LDAP ODBC driver, a BlackBerry Java application, and Outlook, Lotus Notes, and GroupWise connectors.
- Developed embedded C/C++ software for chromatographic device control, contributed to the Stargate build and integration platform, and built an Internet kiosk management system.
- Shipped desktop products (Planet Zip, Planet Split, IChat) in Delphi, C++, and Java, and built and marketed a complete Visual C++ banking solution for private banks.

Open Source Projects

Nex

2026 – Present

Creator and sole author

- github.com/vijaymathew/nex — Contract-first programming language that compiles to the JVM. Built the ANTLR grammar, static type checker, interpreter, ASM bytecode compiler, REPL, LSP server with VS Code and Emacs support, standard library, and cross-platform installers.
- Designed for the AI-coding era: contracts act as executable specifications that LLM-generated implementations must satisfy.

Lumen

Ongoing

Creator

- github.com/vijaymathew/lumen — Fast, non-destructive photo editor for Linux and macOS, built around an immersive canvas, command-palette workflow, and real-time GPU-accelerated editing pipeline.

Slogan and Reo

2018 – 2020

Creator (Slogan), Contributor (Reo)

- Slogan — high-level dynamic programming language that compiles to efficient machine code and supports parallel execution. Reo — programming language and environment for data analysis.

Technical Publications & Writings

- **Books:** *Tight Core, Open Edge* — software design in Nex, with contracts as the membrane between a constrained core and open edges; *Beyond Code* — books and guided tutorials teaching software engineering discipline; *Programming with Nex* — a complete guide to the language.
- **Essays:** *When Code Is Cheap, Design the Promises* — AI has made code cheap to write but no cheaper to trust; how specification and change tracking must adapt. *Designed for Correctness* — why design by contract matters in the age of LLM-generated code.

Education

Mahatma Gandhi University, Kottayam 1995 – 1998

Bachelor's Degree; Major: Financial Accounting with Cost Accounting and Statistics